

Introduction

This report presents a Banking System developed in C++ using Object-Oriented Programming (OOP) concepts. The program allows customers to create accounts, deposit money, withdraw funds, transfer money between accounts, and view transaction history.

Features

Customer Management: Create new customers and accounts.

Deposit: Add money to an account.

Withdraw: Remove money from an account with balance check.

Fund Transfer: Transfer money between two accounts.

Transaction History: View all transactions for an account.

Account Information: Display account number and balance.

Class Design

Transaction: Stores transaction type, amount, balance after, and date.

Account: **Manages** balance and transaction history. Provides deposit, withdraw, and transfer functions.

Customer: Stores customer information and links to an account.

OOP Concepts Used

Encapsulation: Data members are private; access is provided through public functions.

Abstraction: Classes hide complex implementation details.

Constructors/Destructors: Used for initialization and memory cleanup.

Composition: Customer has an Account; Account has Transactions.

Output

```
[OK] Account #5001 created for Iqra Maqsood.
=====
CodeAlpha BANKING SYSTEM
=====
1. Create New Customer
2. Create Account for Customer
3. Deposit
4. Withdraw
5. Transfer Funds
6. View Account Info
7. View Transaction History
8. View All Customers
9. Exit
-----
Enter your choice (1-9): 1
Enter customer name: IQRA MAQSOOD MUGHAL
[OK] Customer created. ID: 1002

Press Enter to continue...
```

Create New Customer

```
C:\Users\iqra0\OneDrive\Documents\codealpha\CodeAlpha_C++ task\task4\banking_system.exe
=====
CodeAlpha BANKING SYSTEM
=====
1. Create New Customer
2. Create Account for Customer
3. Deposit
4. Withdraw
5. Transfer Funds
6. View Account Info
7. View Transaction History
8. View All Customers
9. Exit
-----
Enter your choice (1-9): 2
Enter Customer ID: 1002
Enter initial deposit: $2000
[OK] Account #5002 created for IQRA MAQSOOD MUGHAL.

Press Enter to continue...
```

Create Account for Customer

```
C:\Users\iqra0\OneDrive\Documents\codealpha\CodeAlpha_C++ task\task4\banking_system.exe

=====
CodeAlpha BANKING SYSTEM
=====
1. Create New Customer
2. Create Account for Customer
3. Deposit
4. Withdraw
5. Transfer Funds
6. View Account Info
7. View Transaction History
8. View All Customers
9. Exit
-----
Enter your choice (1-9): 3
Enter Account Number: 5002
Enter Deposit Amount: $2000
[OK] Deposited $2000.00. New balance: $4000.00

Press Enter to continue..._
```

Deposit

```
C:\Users\iqra0\OneDrive\Documents\codealpha\CodeAlpha_C++ task\task4\banking_system.exe

=====
CodeAlpha BANKING SYSTEM
=====
1. Create New Customer
2. Create Account for Customer
3. Deposit
4. Withdraw
5. Transfer Funds
6. View Account Info
7. View Transaction History
8. View All Customers
9. Exit
-----
Enter your choice (1-9): 4
Enter Account Number: 5002
Enter Withdrawal Amount: $100
[OK] Withdrew $100.00. New balance: $3900.00

Press Enter to continue..._
```

Withdraw

```
C:\Users\iqra0\OneDrive\Documents\codealpha\CodeAlpha_C++ task\task4\banking_system.exe

=====
CodeAlpha BANKING SYSTEM
=====
1. Create New Customer
2. Create Account for Customer
3. Deposit
4. Withdraw
5. Transfer Funds
6. View Account Info
7. View Transaction History
8. View All Customers
9. Exit
-----
Enter your choice (1-9): 5
Enter Sender Account Number: 5001
Enter Receiver Account Number: 5002
Enter Transfer Amount: $300
[!] Invalid sender or receiver account.

Press Enter to continue..._
```

Transfer Fund

```
C:\Users\iqra0\OneDrive\Documents\codealpha\CodeAlpha_C++ task\task4\banking_system.exe

=====
CodeAlpha BANKING SYSTEM
=====
1. Create New Customer
2. Create Account for Customer
3. Deposit
4. Withdraw
5. Transfer Funds
6. View Account Info
7. View Transaction History
8. View All Customers
9. Exit
-----
Enter your choice (1-9): 6
Enter Account Number: 5002
Account #5002 | Balance: $3900.00

Press Enter to continue...
```

View Account Info

```

C:\Users\iqra0\OneDrive\Documents\codealpha\CodeAlpha_C++ task\task4\banking_system.exe

=====
CodeAlpha BANKING SYSTEM
=====
1. Create New Customer
2. Create Account for Customer
3. Deposit
4. Withdraw
5. Transfer Funds
6. View Account Info
7. View Transaction History
8. View All Customers
9. Exit
=====
Enter your choice (1-9): 7
Enter Account Number: 5002

=====
Transaction History - Account #5002
=====
Type                Amount        Balance        Date
-----
Opening             2000.00        2000.00        Today
Deposit             2000.00        4000.00        Today
Withdrawal          100.00         3900.00        Today
=====

Press Enter to continue...

```

View Transaction History

```

C:\Users\iqra0\OneDrive\Documents\codealpha\CodeAlpha_C++ task\task4\banking_system.exe

=====
CodeAlpha BANKING SYSTEM
=====
1. Create New Customer
2. Create Account for Customer
3. Deposit
4. Withdraw
5. Transfer Funds
6. View Account Info
7. View Transaction History
8. View All Customers
9. Exit
=====
Enter your choice (1-9): 8

=====
ALL CUSTOMERS
=====
Customer ID: 1001 | Name: Iqra Maqsood
Account #5002 | Balance: $3900.00
-----
Customer ID: 1002 | Name: IQRA MAQSOOD MUGHAL
Account #5002 | Balance: $3900.00
-----

Press Enter to continue...

```

View All Customers

```
C:\Users\iqra0\OneDrive\Documents\codealpha\CodeAlpha_C++ task\task4\banking_system.exe

=====
CodeAlpha BANKING SYSTEM
=====
1. Create New Customer
2. Create Account for Customer
3. Deposit
4. Withdraw
5. Transfer Funds
6. View Account Info
7. View Transaction History
8. View All Customers
9. Exit
=====
Enter your choice (1-9): 9

Thank you for using CodeAlpha Banking System!
=====

-----
Process exited after 701.9 seconds with return value 3221226356
Press any key to continue . . .
```

Exit

Figure 1: Banking System Output

Conclusion

The Banking System demonstrates the use of classes, objects, encapsulation, and composition in C++. It provides a functional banking interface with deposits, withdrawals, transfers, and transaction history.

References

- C++ Documentation.** (2026). Classes and Objects. Retrieved from <https://en.cppreference.com>
- Stroustrup, B.** (2018). The C++ Programming Language. Pearson